

## Exercises - Introduction:

### Text-based Problems

Use the **age\_calc.py** program provided. Run and experiment with the code. In particular, experiment to see what happens when you introduce minor mistakes into **age\_calc.py**, such as spelling mistakes or incorrect indentation.

1. Write a program which uses two variables called **number** and **twice\_number**. The program should prompt the user for a value for **number**, then assign twice this value to **twice\_number**, and then print out both values on the screen. An example run might produce:

```
Enter a value for 'number': 4
The value of 'number' is 4.
The value of 'twice_number' is 8.
```

Can you make sure that you have printed the full-stops at the end of the line?

2. Conditions in **if** statements can be “AND”ed together using the keyword **and**, so that, for example:

```
if another_age > 4 and another_age < 9:
```

means “if **another\_age** is greater than 4 and less than 9”. By using **if** statements along with the keyword **and**, alter **age\_calc.py** so that if **another\_age** works out to be more than 150, the screen output is:

```
Sorry, but you'll probably be dead by [year]!
```

and if **another\_age** works out to be negative, the screen output is:

```
You weren't born in [year].
```

Remember to test the program with various different inputs from the keyboard.

3. (This question is difficult, so don't worry if you don't do it!) Alter your program from question 3 so that it deals with months as well as years, and produces output such as the following:

```
Enter the current year then press RETURN: 2007
Enter the current month (a number from 1 to 12): 10
Enter your current age in years: 47
Enter the month in which you were born (a number from 1 to 12): 5
Enter the year for which you wish to know your age: 2012
Enter the month in this year: 6

Your age in 6/2012: 52 years and 1 month.
```

The program should cope with singulars and plurals properly in the output, e.g. "1 month" but "2 months".

Hints: you will have to introduce extra variables and you may find the following arithmetical operations useful:

| Symbol | Operation      | Example | Value |
|--------|----------------|---------|-------|
| +      | Addition       | 3 + 5   | 8     |
| -      | Subtraction    | 43 - 25 | 18    |
| *      | Multiplication | 4 * 7   | 28    |
| /      | Division       | 9 / 2   | 4     |
| %      | Modulus        | 20 % 6  | 2     |

(Notice that when the division sign "/" is used with two integers, the result will be a float.)